



Hand Gesture Based Wireless AC Light Dimmer

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This project is based on MPU6050 accelerometer sensor and Arduino to wirelessly control the brightness of light. MPU6050 accelerometer senses the tilt positions of your hand in the X-axis and Y-axis and provides digital readings. Arduino Nano reads this data and transmits it using a 433MHz RF module. This data is received at the receiver end and used to control the brightness of light as per the movement of your hand.

You can use this circuit to control the brightness of light as well as the speed of a fan. The circuit is especially useful for a differently-abled person.

Circuit and working

The circuit of hand gesture-based wireless AC light dimmer comprises a transmitter and a receiver.

Transmitter. The circuit diagram of the transmitter is shown in Fig. 1. It has the Arduino Nano board (Board1), 433MHz RF transmitter module (TX1), 9V battery, and MPU6050 accelerometer sensor.

The transmitter module (TX1) has four pins, namely, Vcc, data, ground (GND), and antenna (ANT), as shown in Fig. 1. The Vcc pin can have a wide range of input voltage from 3V to 12V. The transmitter consumes a minimum of 9mA current, which can go as high as 40mA during transmission.

The data to be transmitted is fed to data pin. This signal is then modulated using ASK (amplitude shift keying) and sent through the air at a frequency of 433MHz. The speed at which it transmits data is around 10kbps.

MPU6050 sensor has an integrated 3-axis MEMS (micro electrical mechanical system)

accelerometer and 3-axis MEMS gyroscope. MPU6050 is a 6 DOF (degree of freedom) or 6-axis IMU (inertial measurement unit) sensor. It gives six values in the output—three values from the accelerometer and another three from the gyroscope. This sensor uses the I2C protocol for communication. The accelerometer works on the principle of the piezoelectric effect.

Arduino Nano has thirty pins. Its VIN pin is connected to the positive terminal of the 9V battery. GND pin is connected to the negative terminal of the battery.

MPU6050 is an 8-pin sensor. Its VCC pin is connected to 5V of Arduino Nano. GND pin is connected to the GND pin of Arduino. Pins SCL and SDA of the sensor are connected to pins A5 and A4 of Arduino, respectively. Pin 1 of the TX1 module is connected to GND, pin 2 to pin D11 of Arduino Nano and pin 3 to +9V. Pin 4 (ANT) is connected to an antenna for long-distance transmission.

Software. Arduino IDE is used for

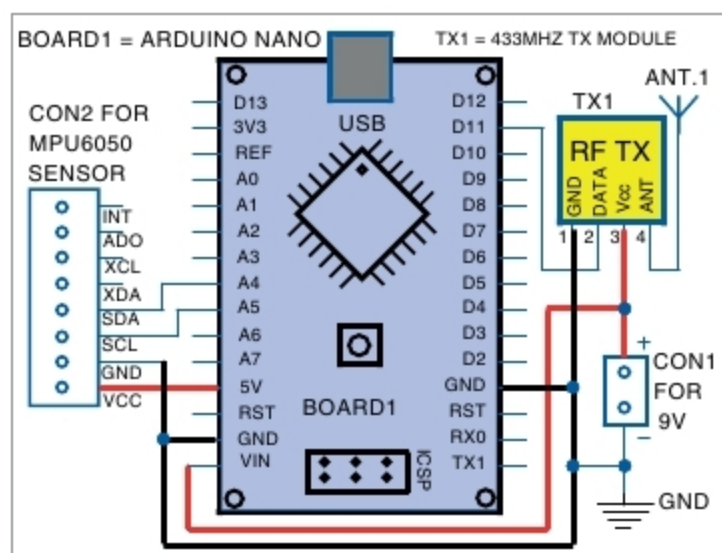


Fig. 1: Transmitter circuit

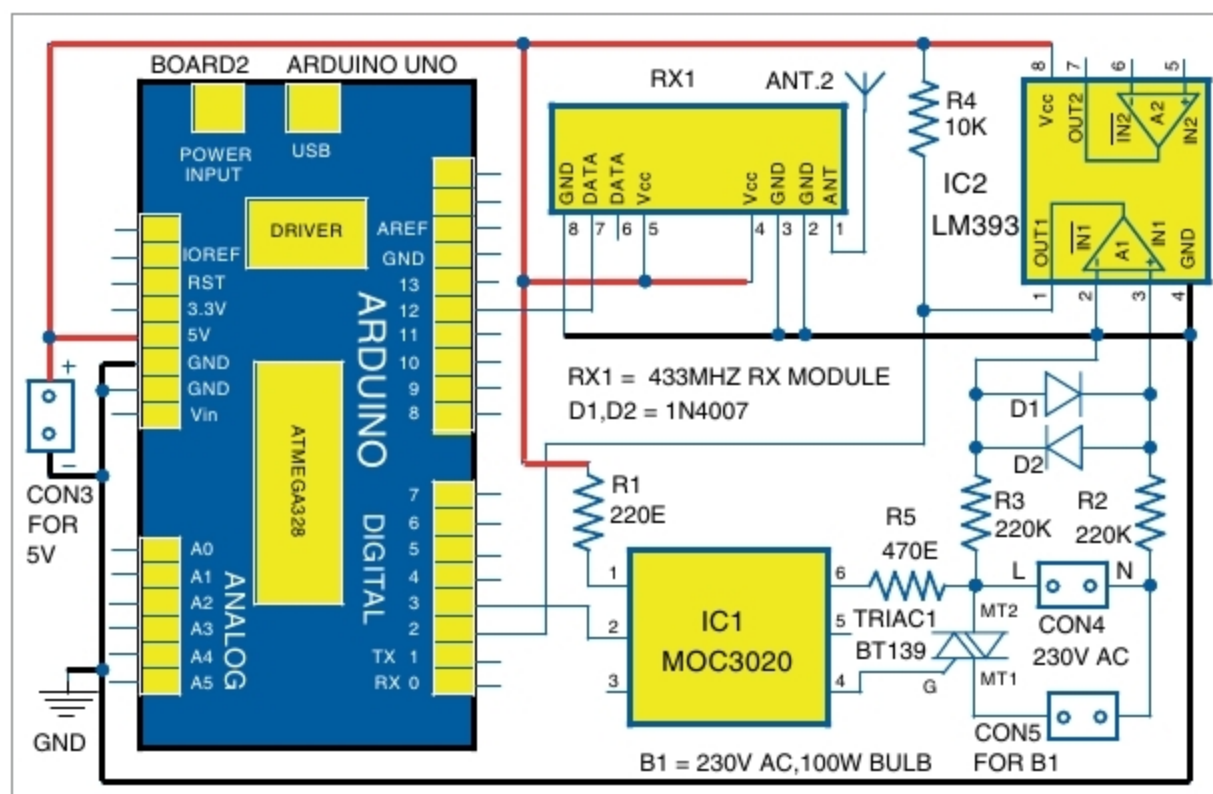


Fig. 2: Receiver circuit

PARTS LIST

Semiconductors:

- Board1 - Arduino Nano
- Board2 - Arduino Uno
- IC1 - MOC3020 opto coupler
- IC2 - LM393 op-amp
- Triac1 - BT139/BTA12 triac
- D1, D2 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 220-ohm
- R2, R3 - 220-kilo-ohm
- R4 - 10-kilo-ohm
- R5 - 470-ohm

Miscellaneous:

- CON1, CON3 - 2-pin connector male and female
- CON2 - 8-pin Bergstrip
- CON4, CON5 - 2-pin terminal connector
- TX1 - 433MHz RF transmitter module
- RX1 - 433MHz RF receiver module
- ANT.1, ANT.2 - 30mm single strand wire
- B1 - 230V AC, 100W bulb
- MPU6050 accelerometer sensor
- 9V battery/9V regulated power supply
- 5V regulated power supply
- Heat sink for BT139